

Experimental data for a liquid/vapor binary mixture of pentane (P) and heptane (H) is given in the table. The quantity z_{liq} gives the mole fraction of pentane in the liquid phase while z_{vap} gives the mole fraction of pentane in the vapor phase. The data can be linearized using the relation $Y = AX$ where A is the separation factor, $X = z_{liq}/(1 - z_{liq})$, and $Y = z_{vap}/(1 - z_{vap})$.

For a general straight line with the intercept equal to zero $y = ax$, the "best fit" slope for perpendicular offsets is given by $a = -\bar{B} \pm \sqrt{\bar{B}^2 + 1}$ where $\bar{B} = \frac{S_{xx} - S_{yy}}{2S_{xy}}$ and the

averages are defined as $S_{xx} = \frac{1}{n} \sum_{i=1}^n x_i^2$, $S_{yy} = \frac{1}{n} \sum_{i=1}^n y_i^2$, and $S_{xy} = \frac{1}{n} \sum_{i=1}^n x_i y_i$, respectively. If $S_{xx} - S_{yy} > 0$, use the plus sign in front of the square root in the expression for the slope. If $S_{xx} - S_{yy} < 0$, use the minus sign.

(a) Give the units on the mole fractions z_{liq} and z_{vap} .

(b) Give the units on the mole fraction ratios X and Y .

(c) Calculate the "best fit" value of the separation factor using perpendicular offsets as described above. Show all of your work.

Mole Fraction z_{liq}	Mole Fraction z_{vap}
0.867	0.984
0.594	0.925
0.398	0.836
0.254	0.701
0.145	0.521
0.059	0.271

$$\frac{(x - S_x)^2}{(y - S_y)^2}$$

(a) $z_{liq} [=] \frac{\text{mol P}}{\text{mol total liq}}$

$z_{vap} [=] \frac{\text{mol P}}{\text{mol total vap}}$

(b) $X [=] \frac{\frac{\text{mol P liq}}{\text{mol total liq}}}{1 - \frac{\text{mol P liq}}{\text{mol total liq}}}$

$\frac{\text{mol H liq}}{\text{mol total liq}} = \frac{\text{mol P liq}}{\text{mol total liq}}$

$Y [=] \frac{\frac{\text{mol P vap}}{\text{mol total vap}}}{1 - \frac{\text{mol P vap}}{\text{mol total vap}}} = \frac{\text{mol H vap}}{\text{mol total vap}} = \frac{\text{mol P vap}}{\text{mol total vap}}$

(c) ok
back

(c)

$\frac{x_{liq}}{(1-x_{liq})}$	$\frac{x_{vap}}{1-x_{vap}}$	$\left(\frac{x_{liq}}{(1-x_{liq})}\right)^2$	$\left(\frac{x_{vap}}{(1-x_{vap})}\right)^2$	$\left(\frac{x_{liq}x_{vap}}{(1-x_{liq})(1-x_{vap})}\right)$
6.519	61.50	42.50	3782.25	400.9
1.463	12.33	2.14	152.0	18.04
0.6611	5.098	.4371	26.00	3.370
0.3405	2.344	.1159	5.494	.7981
.1696	1.088	.0288	1.184	.1845
.06270	.3717	.003931	.1382	.0237

AVG

$$S_{xx} = 7.54 \quad 661.2 = S_{yy} \quad 70.55 = S_{xy}$$

$$\bar{B} = \frac{S_{xx} - S_{yy}}{2S_{xy}} = -4.633 \quad S_{xx} < S_{yy} \rightarrow \text{use minus sign}$$

$$a = -\bar{B} - \sqrt{\bar{B}^2 + 1} \rightarrow a = -(-4.633) - \sqrt{(-4.633)^2 + 1}$$

$$= -0.1067$$

$$y = -0.1067x$$

$$A = -0.1067$$